4 **(a)** Fig. 4.1 shows a liquid-in-glass thermometer labelled thermometer X.

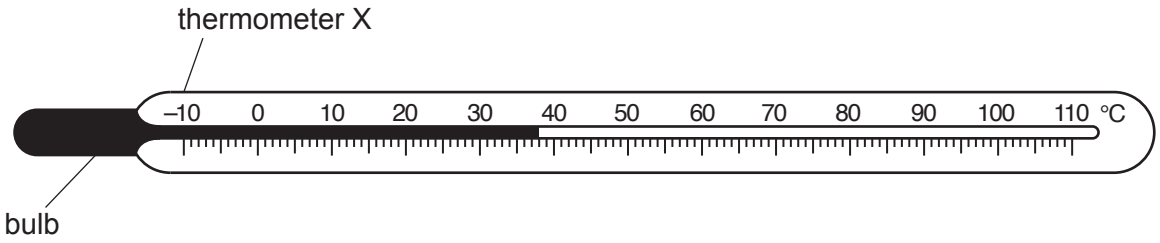


Fig. 4.1

(i) State the physical property which varies with temperature in a liquid-in-glass thermometer.

..... [1]

(ii) Thermometer Y has a bulb that contains twice the volume of liquid compared to thermometer X.

State and explain how the sensitivity of thermometer Y compares with the sensitivity of thermometer X.

statement .....

explanation .....

.....

[2]

(iii) State and explain **one** change that can be made to the design of thermometer X to increase its range.

statement .....

explanation .....

[2]

(b) A liquid-in-glass thermometer cannot measure a temperature of 1300 °C.

State a physical property which varies with temperature in a thermometer which can measure a temperature of 1300 °C.

..... [1]

[Total: 6]

© UCLES 2022